



The Oxford College of Engineering

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

PHASE 2 – FIRST REVIEW

MULTIMODAL DOCUMENT QA ASSISTANT WITH GROUNDED REASONING & HIGHLIGHTING

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PROBLEM STATEMENT



Existing RAG systems mainly depend on extracted text from documents .



Text extraction / OCR can lose or distort layout, tables, diagrams, and visual information .



There is need of Vision based multimodal RAG approach capable of retrieving and reasoning over the original documents .



Modern documents contain texts , image & tables .

MOTIVATION

Future of AI:

Exploring the transition from text –based AI to multimodal vision –enabled AI systems .

Frontier Technology:

Gaining hands on-experience with VLMs, Multimodal retrieval, embeddings and Modern RAG architectures .

Research & Innovation:

Building on recent research such as VisRAG and following a research – driven approach rather than simple application .

Real World Impact:

Applying modern AI to modern documents where accurate information retrieval is valuable .

OBJECTIVES

1

Study & Understand VisRAG:

Analyze the VisRAG architecture and its core components. Understand vision-based retrieval and generation.

2

Develop a Multimodal RAG Pipeline:

Build a vision-based RAG pipeline for maritime technical documents. Explore Vision-Language Models, visual embeddings, and retrieval.

3

Evaluate on Modern Documents:

Test the approach on modern manuals, tables, diagrams, and technical content. Evaluate retrieval and answer-generation performance.

4

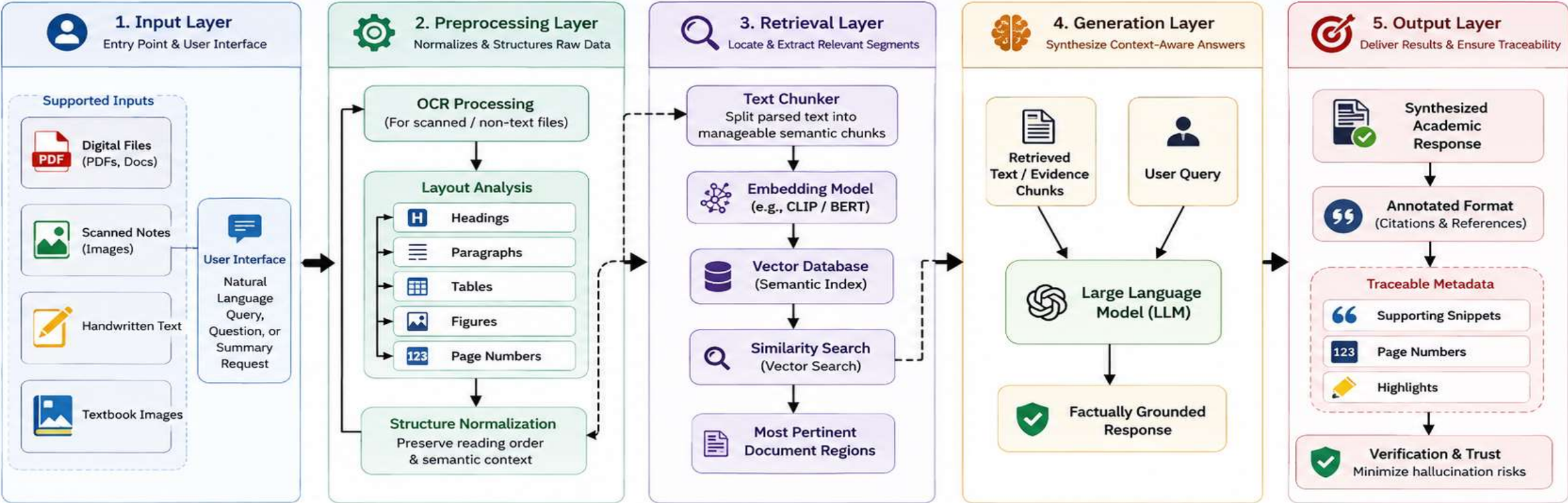
Identify & Improve Limitations:

Analyze system strengths, limitations, and failure cases. Investigate domain-specific improvements for MARIS.

EXISTING SYSTEMS

Multimodal Retrieval-Augmented Generation (RAG) Architecture for Academic Reporting

Unstructured academic and study documents (PDFs, handwritten notes, textbooks) → grounded LLM responses in verified source evidence.
The system **retrieves**, **augments**, and **generates accurate**, traceable, and context-aware academic answers.



Key Implementation Highlights



Multimodal Support
Handles PDFs, images, handwritten notes, textbooks, and more.



Structure-Aware Parsing
Retains document structure (headings, tables, figures, page numbers).



Semantic Retrieval
Uses embeddings & vector search for accurate, context-aware retrieval.



Grounded Generation
LLM answers are grounded in retrieved evidence to ensure factual accuracy.

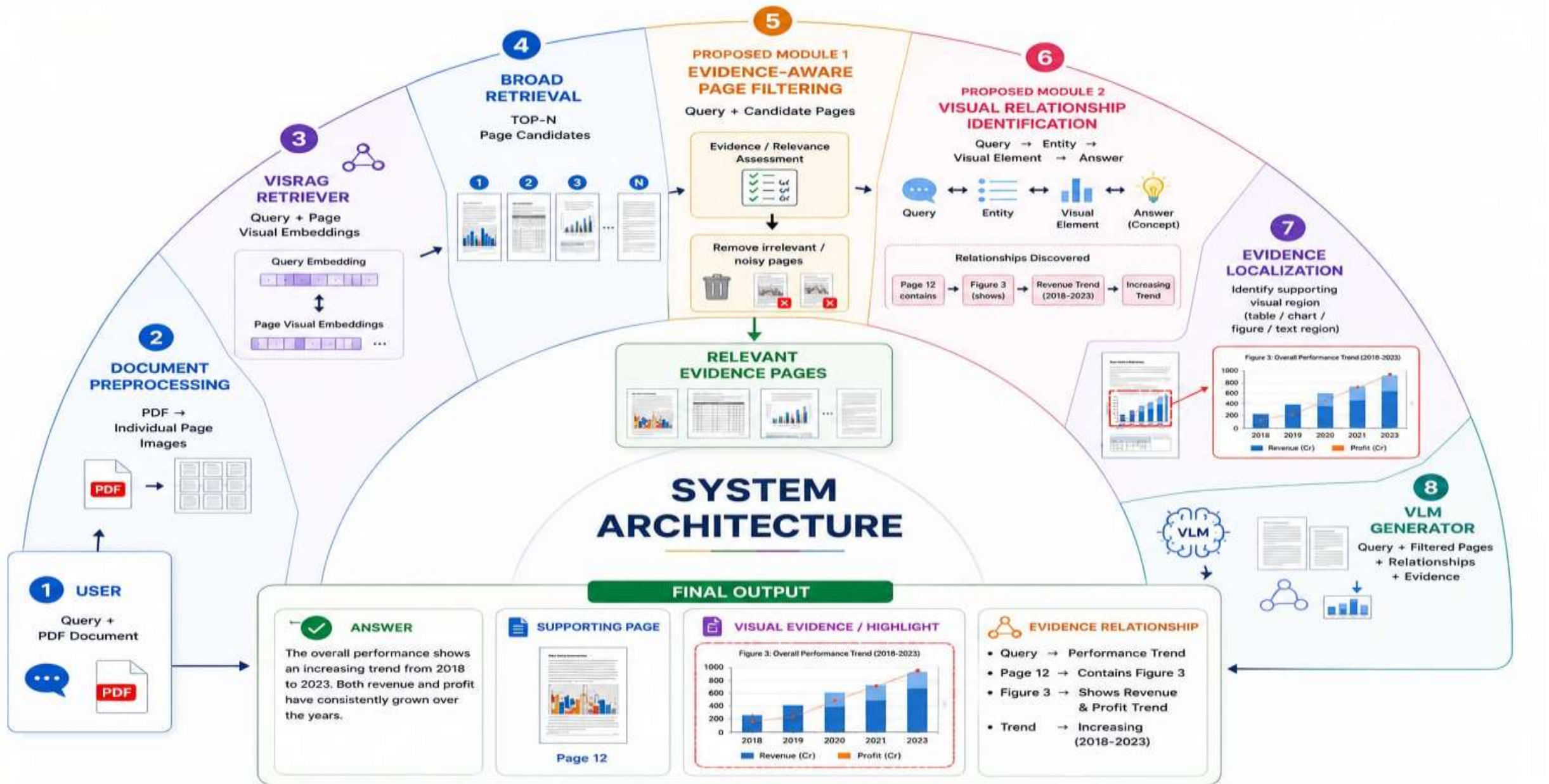


Traceability & Trust
Provides citations, snippets, and page references for transparency and verification.

LITERATURE SURVEY

S.No.	Paper	Authors	Year	Advantage	Research Gap	Result
1	VisRAG: Vision-based Retrieval-Augmented Generation on Multi-Modality Documents	Shi Yu et al.	2023	Direct visual retrieval from document pages; preserves layouts, tables and figures	Generic-domain approach; limited maritime-specific evaluation	Strong visual RAG baseline; selected as MARiS base paper
2	ColPali: Efficient Document Retrieval with Vision Language Models	Manuel Faysse et al.	2024	Effective visual document retrieval using multi-vector representations	Higher storage/compute; mainly retrieval-focused	Improves visual retrieval quality
3	M3DocRAG: Multi-modal Retrieval is What You Need for Multi-page Multi-document Understanding	Jaemin Cho et al.	2023	Handles multi-page and multi-document multimodal understanding	More complex pipeline; domain-specific reasoning remains open	Improves multi-document reasoning and retrieval
4	Vision-Guided Chunking Is All You Need	Vishesh Tripathi et al.	2022	Preserves document structure, tables and figures during chunking	Computational overhead; needs technical-domain validation	Enables structure-aware retrieval
5	Rewrite-Retrieve-Read	Xinbei Ma et al.	2023	Rewrites queries to improve retrieval	Not specifically designed for visual/maritime queries	Improves retrieval for ambiguous queries
6	Adaptive-RAG	Soyeong Jeong et al.	2024	Selects retrieval strategy according to query complexity	Mainly text-based; multimodal adaptation needed	Improves retrieval efficiency and adaptability
7	Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection	Akari Asai et al.	2023	Self-reflection improves factuality and retrieval decisions	Additional inference complexity; visual evidence checking is limited	Improves answer reliability and factuality
8	Corrective Retrieval Augmented Generation (CRAG)	Shi-Qi Yan et al.	2024	Detects poor retrieval and performs corrective retrieval	Visual-page correction remains underexplored	Reduces errors caused by poor retrieval
9	Enabling Large Language Models to Generate Text with Citations (ALCE)	Tianyu Gao et al.	2023	Improves citation quality and answer verifiability	Mostly text-based citations; visual evidence grounding remains open	Produces more verifiable answers
10	RAG-Fusion: a New Take on Retrieval-Augmented Generation	Zackary Rackauckas	2024	Multiple query perspectives and result fusion improve retrieval coverage	Can introduce irrelevant results; not visual-domain specific	Improves retrieval coverage and robustness

SYSTEM ARCHITECTURE



FEATURES IMPLEMENTED

1. Project Development – 60% Progress

Modules Completed

- MARIS system architecture and overall workflow finalized.
- Voice-based interaction system successfully implemented.
- AI tutor conversation flow and interaction design established.
- Document/data ingestion pipeline partially completed.
- Required tools, technologies, models, and resources finalized.

Features Implemented

- Voice input and voice output.
- AI-powered conversational tutoring.
- Core system architecture for integrating ingestion, retrieval, reasoning, and response generation.
- Initial document ingestion capability.
- Foundation for multimodal educational assistance.

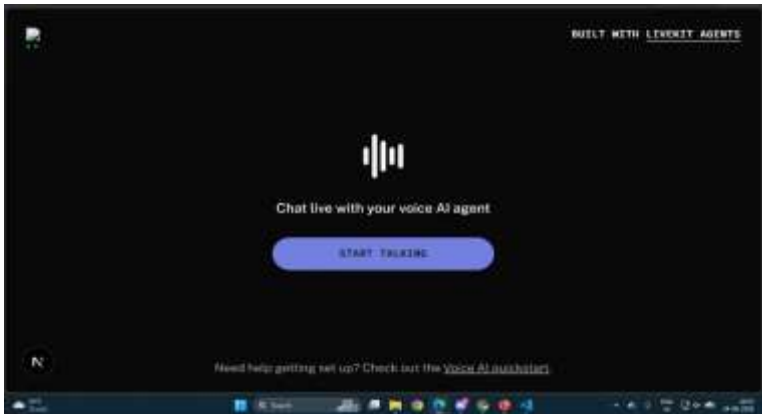
Current Working Demo

- Functional voice-based MARIS prototype.
- User can interact with MARIS through natural voice conversation.
- System architecture is ready for integration of the remaining educational intelligence modules.

REMAINING WORK

Remaining Work – 40%

- Complete the remaining **60% of the ingestion pipeline**.
- Implement document understanding and retrieval.
- Integrate evidence-based question answering.
- Add personalized/adaptive tutoring capabilities.
- Integrate multimodal educational content processing.
- Complete end-to-end MARIS pipeline integration.
- Conduct functional, performance, and user testing.
- Optimize response accuracy, latency, and reliability.
- Prepare the final standalone/demo-ready MARIS system and documentation.



PROJECT TIMELINE



Timeline:

Problem Identification → Literature Survey → System Design → AI Development → 60% Application

Development → 40% Remaining Work → Testing → Final Submission

FUTURE WORK & SCOPE

Improve Response Accuracy – Enhance MARIS’s understanding, reasoning, and response quality across different learning queries.

Multimodal Learning Support – Enable MARIS to work with text, images, documents, and other educational content.

Advanced Voice Interaction – Improve natural conversation, interruption handling, follow-up questions, and context-aware dialogue.

Personalized Learning – Provide adaptive explanations, difficulty levels, learning preferences, and individualized tutoring.

Standalone Application – Package MARIS as an easy-to-use application for students and educators.

User Testing – Evaluate MARIS with different users, subjects, learning levels, and real-world educational scenarios.

Accessibility Expansion – Extend MARIS to support learners with different accessibility needs through voice-first and multimodal interaction.

CONCLUSION



- Developed an evidence-grounded multimodal document QA system that goes beyond simple page retrieval.



- Implemented the VISRAG baseline and used it as the foundation for our proposed approach.



- Added Evidence-Aware Page Filtering to reduce irrelevant and noisy retrieved pages.



- Introduced Visual Relationship Identification to connect the query with relevant entities, charts, tables, figures, and values.



- Added evidence localization to identify the specific region that supports the generated answer.



- Combined retrieval, visual reasoning, and VLM-based generation to produce more transparent and trustworthy answers.



- Used pretrained models and existing APIs to reduce development complexity and focus on the core research contribution.



- The overall approach provides a foundation for more reliable, explainable, and evidence-backed multimodal document understanding.

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THANK YOU !!!